Radiotherapy

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1 Introduction

According to the Worldwide Cancer Research Foundation [2], around 50% of all cancer patients require some form of radiation therapy (radiotherapy) as treatment for the best chance of getting rid of their cancer. Radiotherapy is the technique of aiming high energy beams such as x-rays, protons, or gamma rays to damage specific cancerous cells inside the body to prevent them from reproducing. This allows for the body's healthy cells to slowly take back the cancerous sections as they die out, leading to radiotherapy being wholly or partly responsible for around 40% of cases where a patient's cancer was reportedly cured [2].

Due to the uniqueness in both every patient's body shape and the shape of the cancerous mass, each radiotherapy treatment needs to be calculated and tuned to achieve the desired results. The parameters for a successful treatment includes several dosage requirements at different areas in the body. A minimum dosage is required in units of Gy for the targeted cancerous areas along with a maximum dosage for those same target areas as the goal of radiotherapy is not to burn out the cancer instantly but rather just enough to damage its reproduction ability. The areas around where the cancer is located can often be comprised of critical organs and other bodily tissues with the critical areas being given a maximum safe dosage and the normal tissues having a relatively higher safe dosage in comparison. These different dosages are achieved by firing these rays from a variety of different angles and at a variety of different strength so that the dosage accumulates in the correct areas while staying low in others.

The rays themselves can be modeled in several different ways through methods such as a convolution model, full simulation model, and pencil beam model. For simplicity and to be more time efficient, the pencil beam model will be explored further in this report. The main weakness of the pencil beam model is its inability to accurately represent how the dosage propagates through different tissues which often have different electron density. While this makes it unsuitable for treatments where things like bones may get in the way, it is still highly suitable for other areas of the body where the density is relatively uniform such as the brain.

2 Mathematical Model

The method and model outline used in this study was based on the approach outlined by Holder and Eichholz [1].

To start with accurately representing the pencil beam model, it must first be determined how a single pencil delivers its associated dose into the assumed uniform density tissue. These beams attenuate as they pass through the patient leading to higher dosages near where they first enter the tissue and lower dosages as the beam travels and loses its energy. The beams also deliver this dosage throughout their entire radius with a quick drop off further outside of that. These two beam properties can be modeled by the following equations and beam properties.

$$D_{p,a,i} = \begin{cases} \left(P_0 e^{-\mu(d-1.5}(1 - e^{-\gamma r} + \frac{r d \alpha_d}{r+1.5}\right) \cdot \text{ISF} \cdot \text{O}, & d \geq 1.5 \\ (0.4\frac{d}{1.5} + 0.6)\left(P_0(1 - e^{-\gamma r}) + \frac{1.5 r \alpha_d}{r+1.5}\right) \cdot \text{ISF} \cdot \text{O}, & 0 \leq d < 1.5. \end{cases}$$

The above equation represents attenuation along the axis of the beam. $D_{p,a,i}$ represents the resulting dose. P_0 is dependent on the exact machine used for radiotherapy but was experimentally determined to have a value of 0.873 for this model. The variable μ is an energy parameter and was experimentally determined to be 0.0469 for this model. The γ variable also needs to be experimentally determined and a value of 5.2586 was used for this model. The beam itself is assumed to be square with a side length of 3 mm, so r is initially assumed to be 1.5 mm. To better compensate for the square beam not having a true radius, r is replaced by $1.122r$ to better account for the corners of the square beam. This makes the model's value of r equal to a constant 1.6830. To account for scattered dosage caused by nearby pencils, the $rd\alpha_d/(r + 1.5)$ term was added where $\alpha_d = 0.1299 - 0.0306 \ln(d)$. The final two factors, ISF and O, are the inverse square factor and off axis factor respectively. The ISF is defined as the square of the ratio between the distance from the beam making gantry to the isocenter (l_{gc}), and the distance from the gantry to the point where the dosage is being calculated (l_{gp}). For this model and radiotherapy in general, l_{gc} is set to be 100 cm while l_{gp} depends on the patient's geometry. The ISF equation is stated explicitly below.

$$\text{ISF} = \left(\frac{l_{gc}}{l_{gp}}\right)^2.$$

The off axis factor O is not directly calculated but rather gotten through an experimentally determined lookup table which relates a new variable (ψ) to the factor O. This ψ can be calculated as shown below.

$$\psi = \frac{o \cdot l_{gc}}{s + d}$$

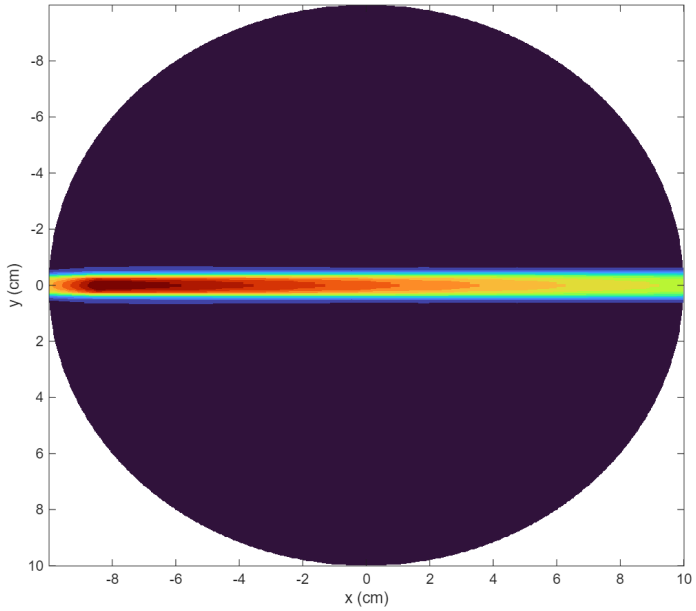


Figure 1: Single Beam Attenuation in Circular Patient

Throughout all these equations, the point for which the dose is being calculated is represented by the coordinates (d, o) where d is the depth into the patient along the beam's axis measured from the point the beam first enters the patient and o is the orthogonal distance from the beam's axis. The variable s is then representative of the distance from the gantry to the surface of the patient.

As a proof that the single beam attenuation model is working correctly, the above figure 1 was created showing the dosage distribution through tissue. From the above equation, it is expected that the maximum dosage seen will be both along the beam's center axis and at a distance of 1.5 cm into the patient. A quick fall off is expected in the off axis direction while a much slower fall off is expected in the direction of the beam. To simplify both the geometry and the math used in the model, the patient and associated tissue was assumed to be circular.

The next steps in properly configuring the model involves creating a way for these beams to come from different angle and be spread out throughout the patient. The gantry where the beams originate from can be orientated anywhere around the patient. The number of beams shot at the target can be configured for each patient's case but in general, the combination of all beams from a single gantry angle needs to be able to approximate the dosage that would result from a single extra large beam. Both of these aspects of the model can be seen in the following figure 2.

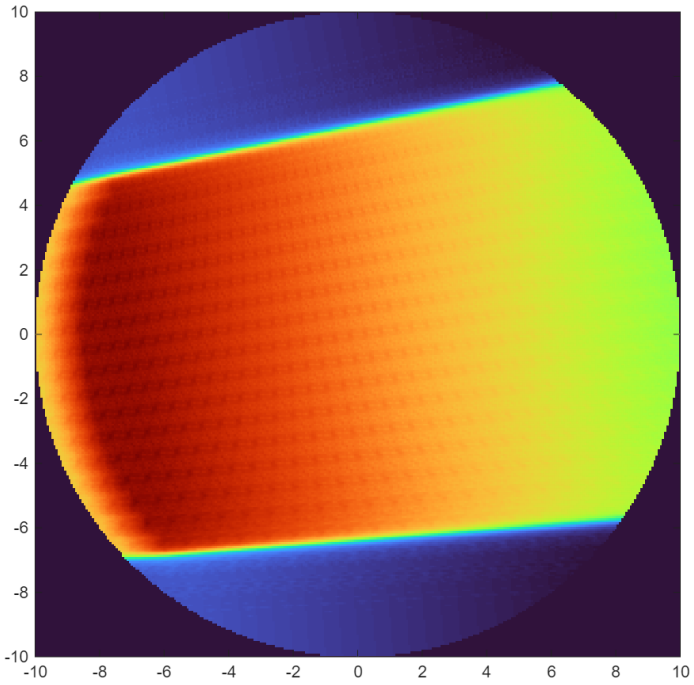


Figure 2: Multi Beam Conglomerate at Angle

3 Optimization

In order to transform the model so far from a group of beams blasting the entirety of the patient from all angles to only involving the beams that strike the target while leaving the vulnerable organs, a fluence map must be determined which will be referred to by the variable x. By combining this fluence map with the effects of all possible beams on the patient, the optimal treatment can be determined. The effects of the beams on the patient will be stored in a matrix D and the fluence map x will be a vector of values containing the strengths of each beam making the solution Dx. The strength of each beam can vary greatly depending on the target dose to control the amount of Gy each beam produces but a majority of the beams will have a strength of 0 which effectively turn off the specified beams.

To start, the patient geometry must be separated into target regions, critical organs, and normal tissue. As no real life data was found due to the sensitive nature of patient's cancer scans, made up data sets and shapes were created and calculated with using MATLAB's inpolygon function. The width of the combined beam shot shown above in figure 2 must also be wide enough to appropriately shoot through the target areas from any angle. An Example of this can be seen in figure 3 where the magenta outline encloses the target tissue and the cyan tissue encloses the protected organs. Everything else inside the patient is assumed to be normal tissue.

Treatment design for this model was comprised of target dosages and limits to optimize around with various weights as penalties for violating the intended dosage. Inside the target tissue, a hard minimum of 70 Gy was enforced to ensure effective treatment while an upper soft limit of 100 Gy was put in place to limit cell damage. It was assumed that the protected organs should aim to receive no more than 45 Gy and a dosage of 25 Gy or less was determined to be inconsequential. The normal tissue was not as stringent with only applying a penalty if any 1 cm by 1 cm section had a dosage over 100 Gy.

In addition to these base parameters, two more custom parameters were introduced to further fine tune solutions. The first was a very slight penalty applied based on the total amount of Gy in the entire patient to reduce the number of unnecessary beams used. The second was a small penalty based on the average dose seen in the normal tissue to attempt to get rid of stray beams that barely hit the target or miss entirely.

Weights for each of these variables are entirely arbitrary and can be fine tuned on a case by case basis for better solutions but the following weights were used for the specific target and organ geometry shown in figure 3. These weights are also normalized with respect to the number of points inside of then according to the following.

$$\lambda_T = \frac{2000}{|T|}, \lambda_C = \frac{1900}{|C|}, \lambda_N = \frac{1800}{|N|}, \lambda_{\text{All}} = \frac{100}{|T| + |C| + |N|}, \lambda_{N_{\text{avg}}} \frac{300}{|N|}$$

The weights themselves are fairly large values instead of being easier values such as 2, 1.9, 1.8, etc, due to the massive number of target points that appear

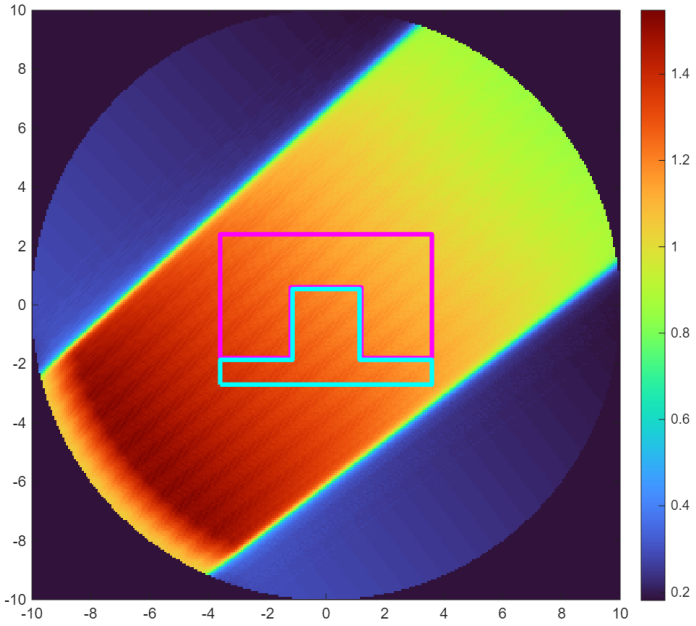


Figure 3: Demonstration of Target and Protected Geometry

within each section seen in the denominator of these weights. These weights combine into the function that will be minimized to determine the optimal solution, shown in the following.

$$\left.\begin{array}{l} \text{minimize } \lambda_T \sum_i \gamma_i + \lambda_C \sum_i \rho_i + \lambda_{N_{\text{avg}}} \sum_i \sigma_i + \lambda_{\text{all}} \sum_i (Dx)_i \\ \text{such that} \\ P_T \leq D_T x \leq P'_T + \gamma \\ D_C x \leq P_C + \rho \\ x \geq 0 \\ \gamma \geq 0 \\ \rho \geq -\eta P_C \\ \sigma \geq 0 \end{array}\right\}$$

It can be seen that the normal tissue hot spot correction weight was left out of the above equations. This is due to both it being replaced by the two custom weights, and that by removing it, the optimization is able to solve faster while still being of exceptional quality.

Of the remaining variables, P_T represents the target dose, P'_T represents the upper limit to the target dose, P_C represents the upper dose for at risk organs, and η represents the value for which no penalty is applied to dosage in the at risk organs or 25/45 for this specific case. The variables $\gamma, \rho,$ and σ are used to represent tissue only present at the target, protected organ, or normal tissue locations respectively.

4 Results

The model was made using MATLAB and is able to output a variety of different aspects of the solution based on what the user desires and what settings are applied. The solve time can vary greatly based on parameters such as the solution resolution and step size between gantry angles. The primary output of the model is two figures, a mapping of the applied dosage in the patient and a dose-volume histogram for the associated solution.

As with all previous figures, the patient was assumed to be a perfect circle, 10 cm in radius. The attached colorbar shows the dosage in Gy spread throughout the patient. The small red outlined areas in the normal tissue is showing where the 1 cm by 1 cm hot spots have been detected. These hot spots do not disqualify the solution, they merely indicate that tissue in those areas may die or experience adverse effects. In reality, a physician would compare the dosage map to the patient's actual geometry to ensure the patient's safety.

The histogram allows for the user to see how much of the target, protected organs, or normal tissue receives what percent of the target dose. Ideally, the organs and normal tissue would quickly decline to a low percent target dose while the target tissue receives the full dosage throughout its volume then sharply declines to indicate a low amount of overshoot has occurred. The high amount of overlap in this specific case is typical and exaggerated by the small gaps between target, organ, and normal tissue for better visualization.

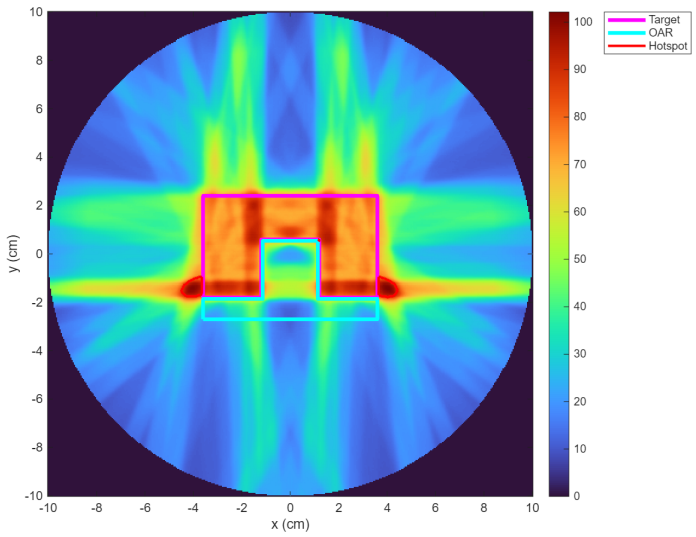


Figure 4: Optimized Treatment for Made Up Cancer Case

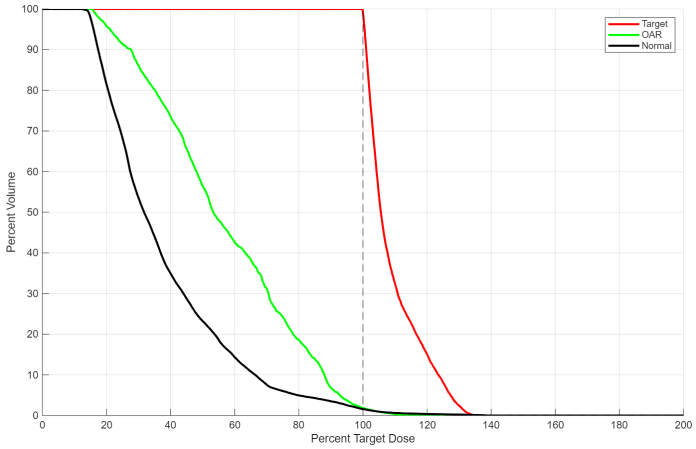


Figure 5: Histogram for Figure 4 Solution Map

5 Conclusion

This model turned out to be very successful in its intended goal of making treatment mappings. In addition to the cancer geometry example shown in this report, several more were experimented with but not tuned as finely and exceptional results were produced every time. Future iterations of this model may experiment with several different optimization options to determine if even better results can be achieved. Some possible avenues of further research include switching the objective function to be based on the maximum found instead of the area as a whole, replacing the hard limit for target dosage with a soft limit and large weight to be fine tuned, and switching the objective function to a quadratic model, using MATLAB's quadprog instead of linprog.

References

[1] Allen Holder and Joseph Eichholz. *An Introduction to Computational Science*, volume 278. Springer Nature Switzerland AG, 2019.

[2] Worldwide Cancer Research. How effective is radiotherapy?, August 2025. Accessed: 2026-03-23.